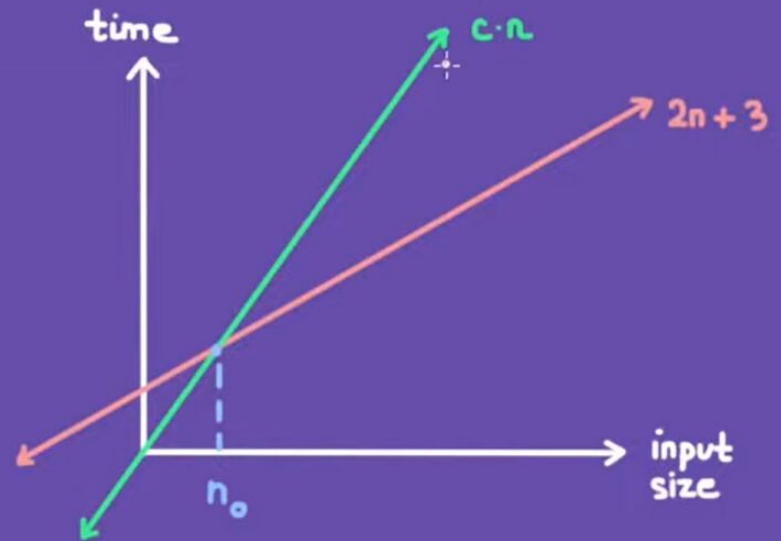


Prove that $2n + 3$ is $O(n)$

$$2n + 3 \leq 2n + 3n$$

$$2n + 3 \leq 5n$$

So, let $c = 5$



Prove that $2n + 3$ is $O(n)$

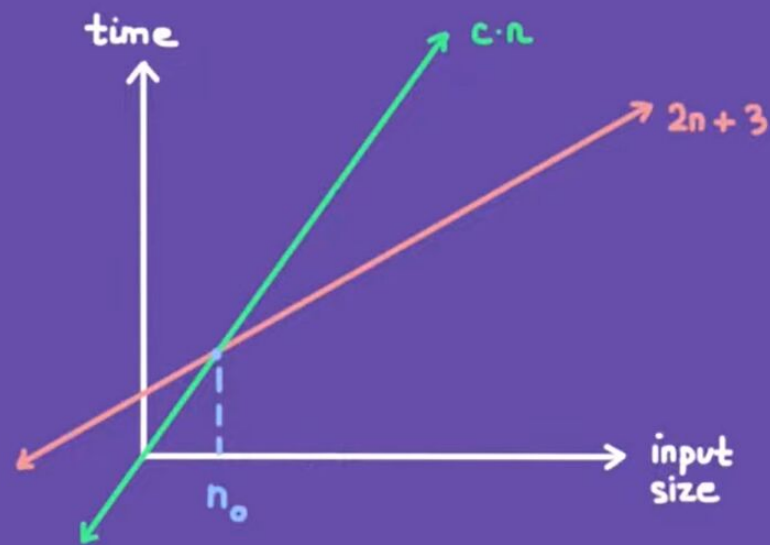
$$2n + 3 \leq 2n + 3n$$

$$2n + 3 \leq 5n$$

$$n_0 = 1$$

$$2(1) + 3 \leq 5(1)$$

$$5 \leq 5 \checkmark$$



Prove that $2n + 3$ is $O(n)$

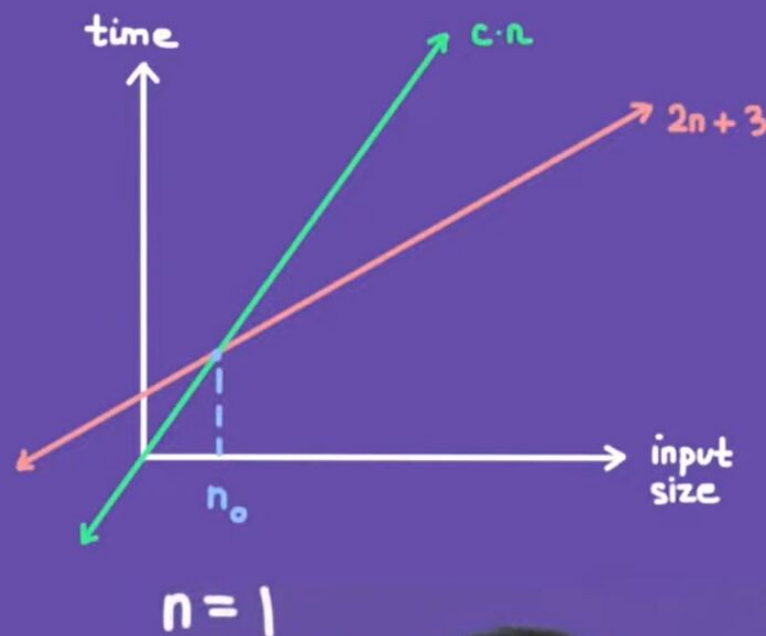
$$2n + 3 \leq 2n + 3n$$

$$2n + 3 \leq 5n$$

By definition, with $c=5$ and $n_0=1$,

Therefore, $2n+3$ is $O(n)$ since

$$2n + 3 \leq 5n \text{ for } n \geq 1.$$

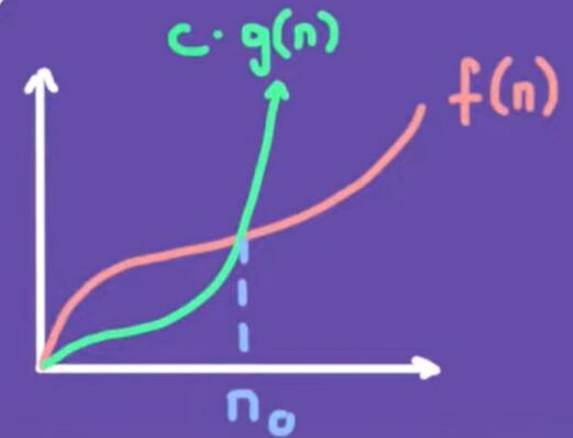


Definition of Big O

$f(n)$ is $O(g(n))$

If there is a positive constant c
and a positive integer n_0

such that $0 \leq f(n) \leq c \cdot g(n)$
for all $n \geq n_0$

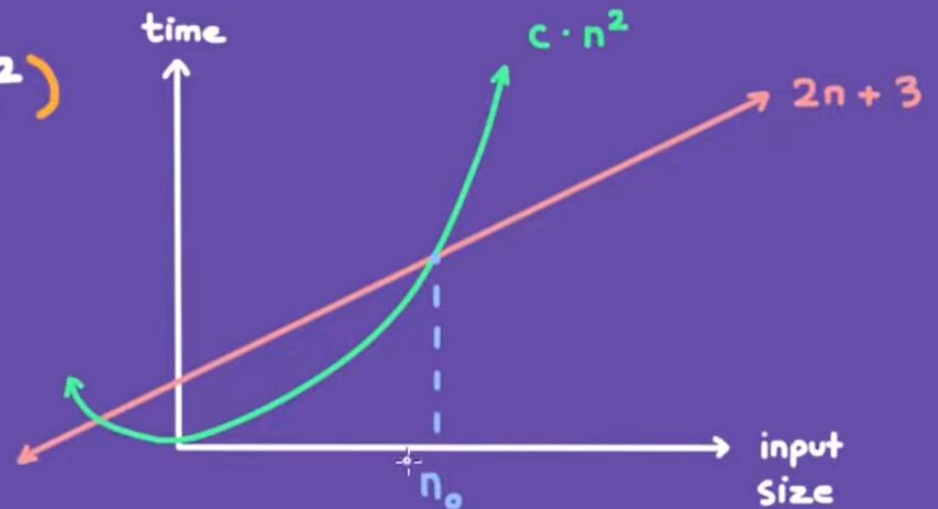


Prove that $2n + 3$ is $O(n^2)$

$$2n + 3 \leq 2n^2 + 3n^2$$

$$2n + 3 \leq 5 \cdot n^2$$

So, let $c = 5$

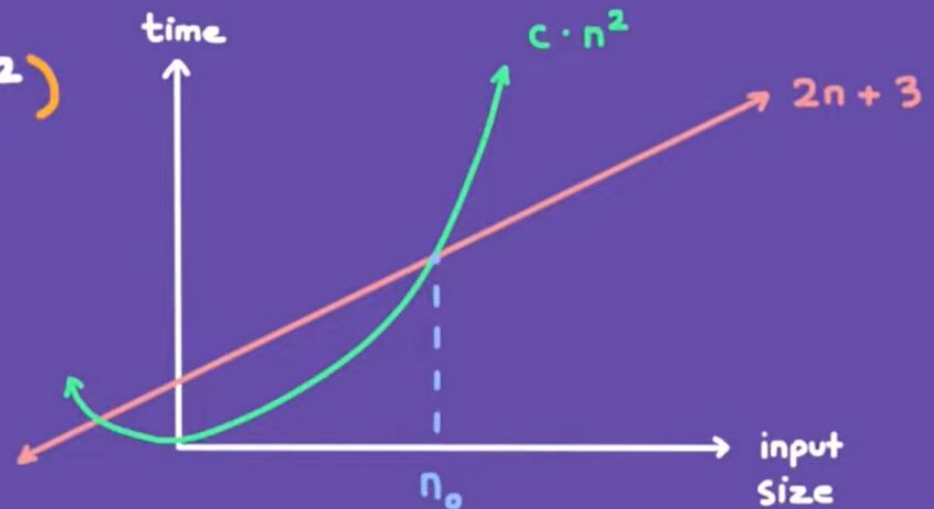


Prove that $2n + 3$ is $O(n^2)$

$$2n + 3 \leq 2n^2 + 3n^2$$

$$2n + 3 \leq 5 \cdot n^2$$

for all $n \geq 1$



By definition of Big O, with $c=5$ and $n_0=1$,

Therefore, $2n + 3$ is $O(n^2)$ since

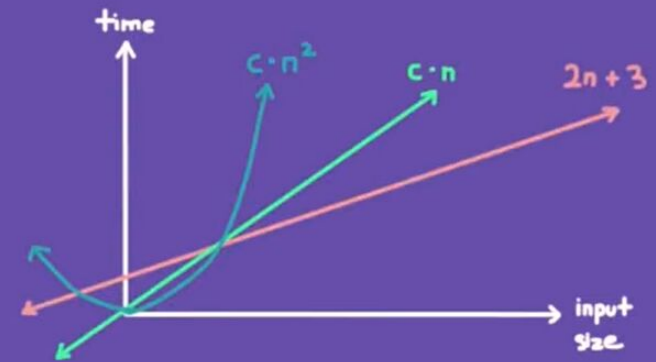
$$2n + 3 \leq 5n^2 \text{ for } n \geq 1$$



Which is the tightest Big O ?

a) $2n + 3$ is $O(n^2)$

b) $2n + 3$ is $O(n)$



What is the tightest big O of this function?

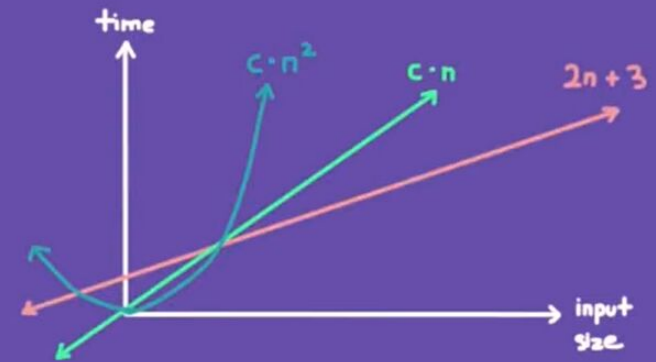
$$7n^3 + 8\log n + 9n^2 + 11$$



Which is the tightest Big O ?

a) $2n + 3$ is $O(n^2)$

b) $2n + 3$ is $O(n)$



What is the tightest big O of this function?

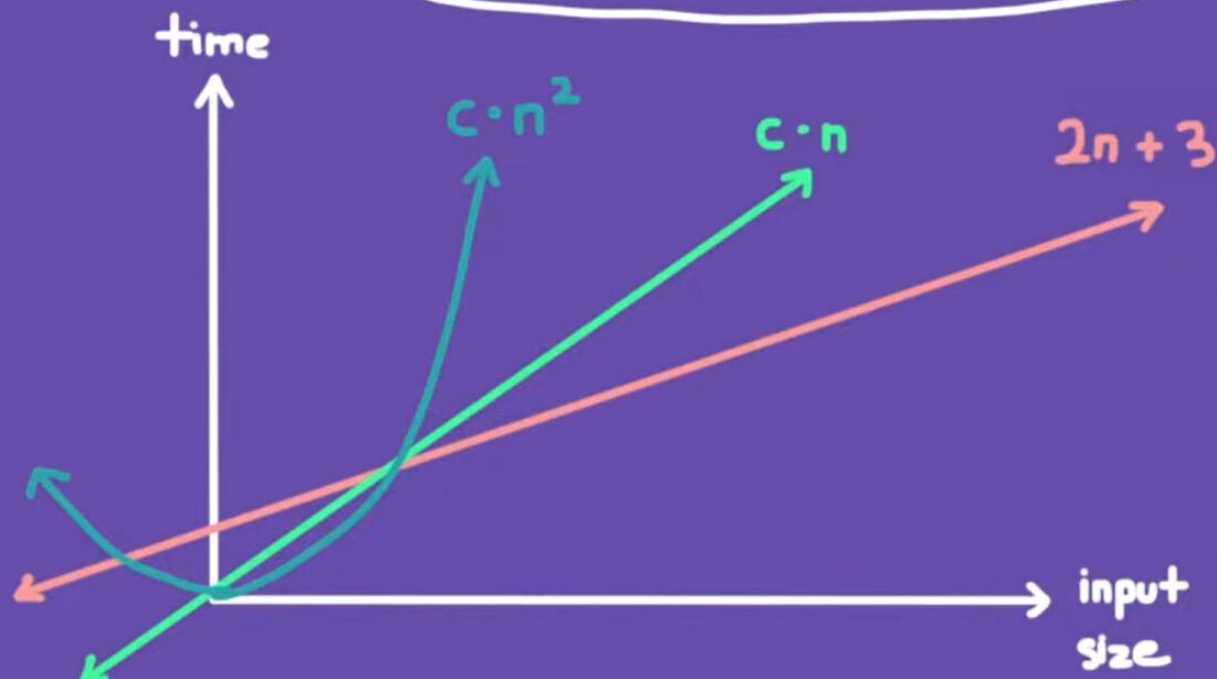
$$7n^3 + 8\log n + 9n^2 + 11$$



Which is the tightest Big O ?

a) $2n + 3$ is $O(n^2)$

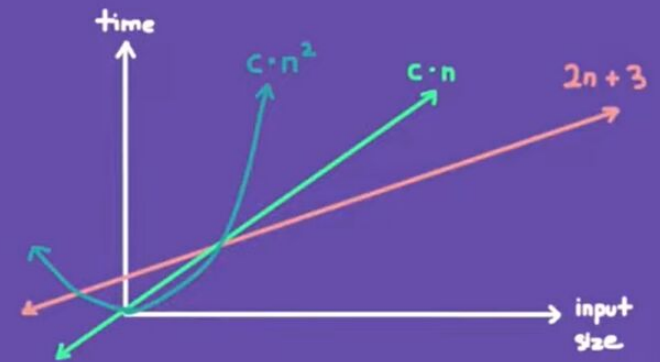
b) $2n + 3$ is $O(n)$



Which is the tightest Big O ?

a) $2n + 3$ is $O(n^2)$

b) $2n + 3$ is $O(n)$



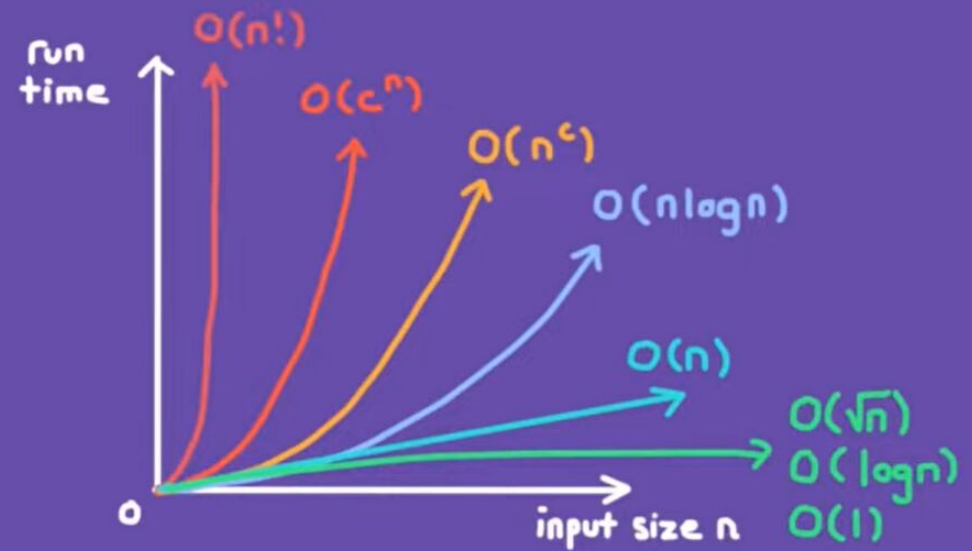
$O(n^3)$

$13 \rightarrow O(1)$

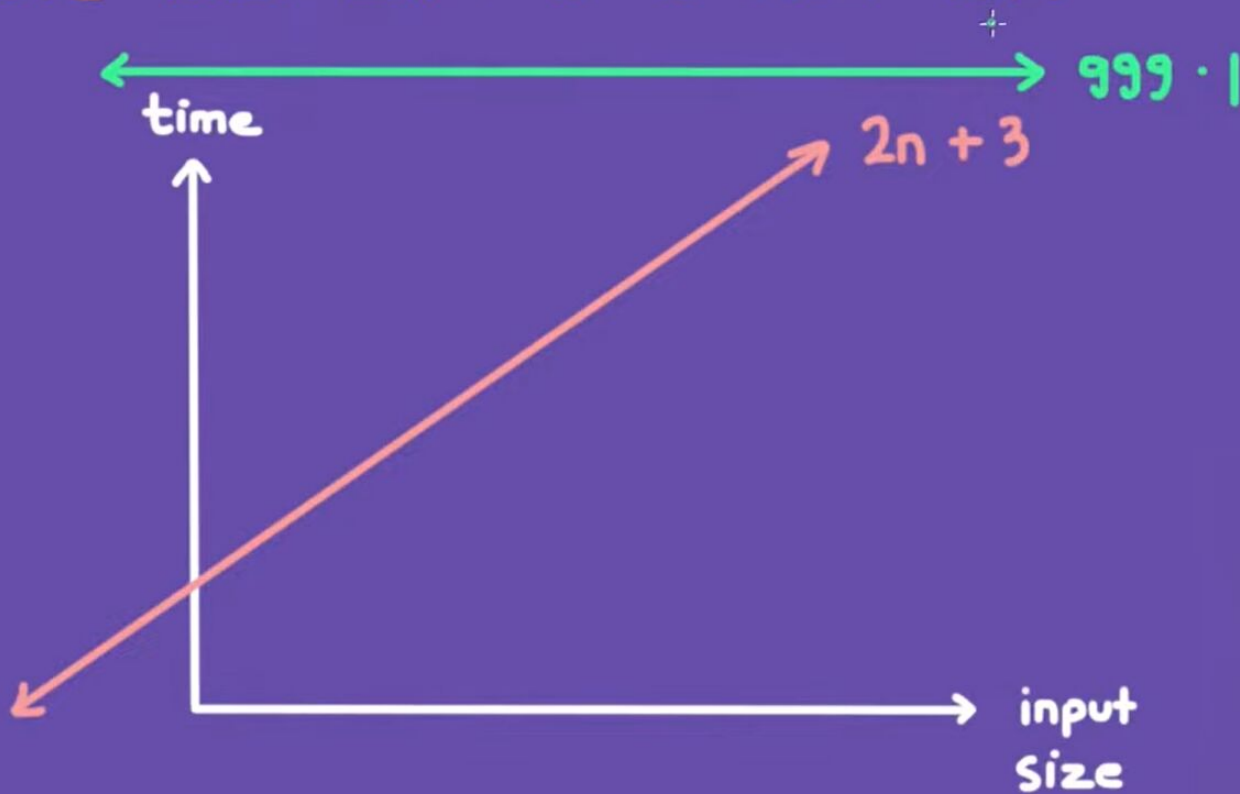


n^n
 $+$ $n!$
 c^n
 n^c
 $n \log n$
 n
 $\sqrt{n}$
 $\log n$
 c

Nowadays
 Fascinating
 Experiments
 Push
 New
 Limits :
 Scientific
 Learning &
 Creativity



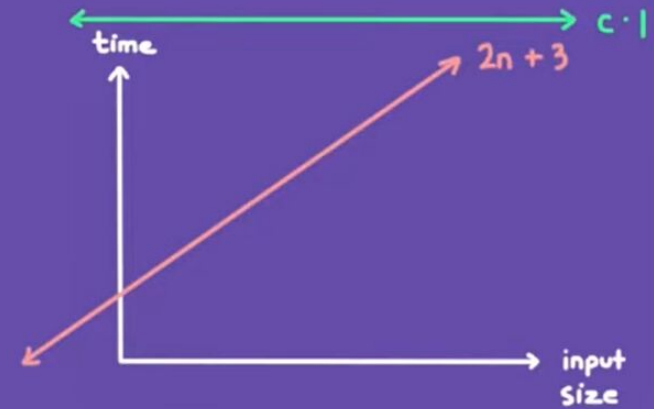
Prove that $2n + 3$ is NOT $O(1)$



Prove that $2n + 3$ is NOT $O(1)$

Prove p by Contradiction:

1. Suppose $\bar{p}$
2. Show that $\bar{p}$ leads to a contradiction



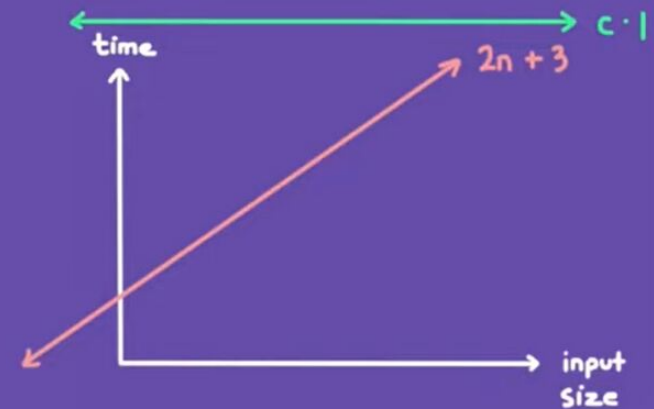
Prove that $2n + 3$ is NOT $O(1)$

E.g : Prove that a box containing
3 cookies is not empty



1. Suppose the box is empty
2. This means you won't pull anything out of the box.

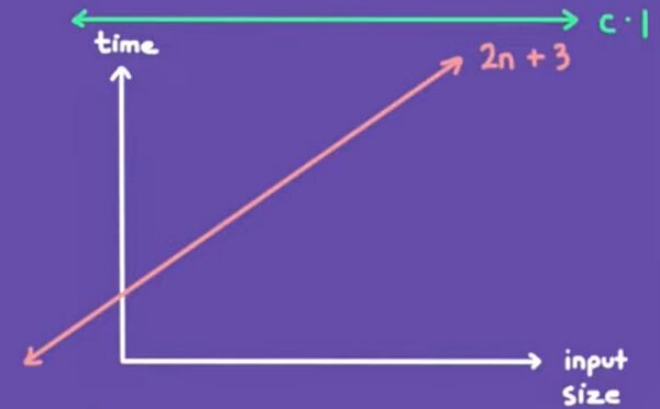
But when I reach into the box, I managed to pull out a cookie.



Prove that $2n + 3$ is NOT $O(1)$

Suppose $2n + 3$ is $O(1)$

Then, there exists a positive constant c and a positive integer n_0 such that $2n + 3 \leq c \cdot 1$ for $n \geq n_0$



This is a contradiction as $2n + 3$ grows infinitely so it cannot be bounded by a constant. Thus, $2n + 3$ is not $O(1)$

+



Prove that 2^{n+2} is $O(2^n)$

$$2^{n+2} = (2^n)(2^2) = (4)2^n$$

$$2^{n+2} \leq 4 \cdot 2^n \text{ for } n \geq 1$$

let $n_0 = 1$

$$2^3 \leq 4 \cdot 2$$

$$8 \leq 8 \checkmark$$

$f(n)$ is $O(g(n))$

If there is a positive constant c
and a positive integer n_0

such that $0 \leq f(n) \leq c \cdot g(n)$
for all $n \geq n_0$



Prove that 2^{n+2} is $O(2^n)$

$$2^{n+2} = (2^n)(2^2) = (4)2^n$$

$$2^{n+2} \leq 4 \cdot 2^n \text{ for } n \geq 1$$

By the definition of Big O, with $c=4$ and $n_0=1$

Therefore, 2^{n+2} is $O(2^n)$ since
 $2^{n+2} \leq 4 \cdot 2^n$ for $n \geq 1$

$f(n)$ is $O(g(n))$

If there is a positive constant c
and a positive integer n_0

such that $0 \leq f(n) \leq c \cdot g(n)$
for all $n \geq n_0$



Prove that

$5n^2 + 3n \log n + 2n + 5$ is $O(n^2)$

$f(n)$ is $O(g(n))$

If there is a positive constant c
and a positive integer n_0

such that $0 \leq f(n) \leq c \cdot g(n)$
for all $n \geq n_0$



Prove that

$5n^2 + 3n \log n + 2n + 5$ is $O(n^2)$

$f(n)$ is $O(g(n))$

If there is a positive constant c
and a positive integer n_0

such that $0 \leq f(n) \leq c \cdot g(n)$
for all $n \geq n_0$

$$5n^2 + 3n \log n + 2n + 5 \leq 5n^2 + 3n^2 + 2n^2 + 5n^2$$

$$5n^2 + 3n \log n + 2n + 5 \leq 15n^2$$

So, let $c = 15$



Prove that

$5n^2 + 3n \log n + 2n + 5$ is $O(n^2)$

$f(n)$ is $O(g(n))$

If there is a positive constant c
and a positive integer n_0

such that $0 \leq f(n) \leq c \cdot g(n)$
for all $n \geq n_0$

$$5n^2 + 3n \log n + 2n + 5 \leq 5n^2 + 3n^2 + 2n^2 + 5n^2$$

$$5n^2 + 3n \log n + 2n + 5 \leq 15n^2$$

$$\text{let } n_0 = 1$$

$$5 + 0 + 2 + 5 \leq 15$$

$$12 \leq 15$$



Prove that

$5n^2 + 3n \log n + 2n + 5$ is $O(n^2)$

c^n

n^c

$n \log n$

$$5n^2 + 3n \log n + 2n + 5 \leq 5n^2 + 3n^2 + 2n + 5$$

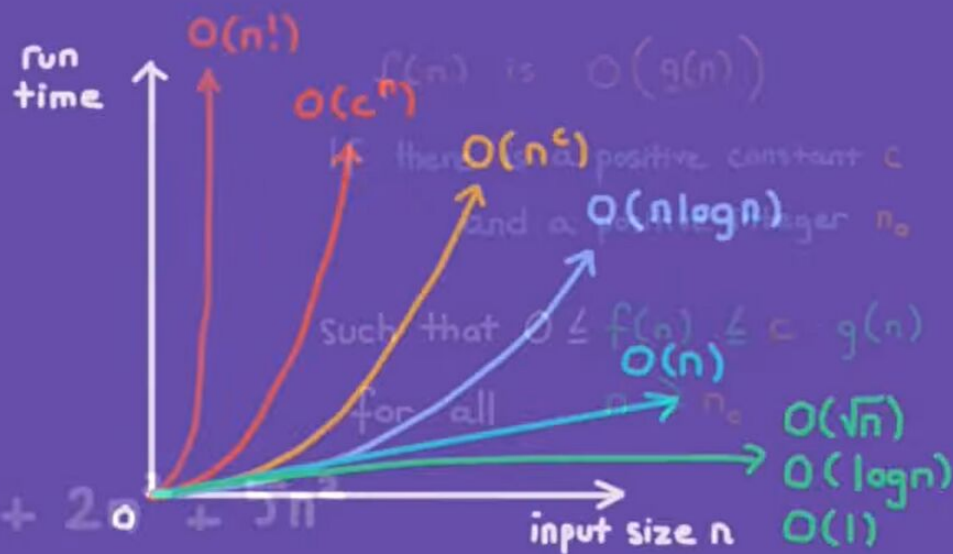
$$5n^2 + 3n \log n + 2n + 5 \leq 15n^2$$

$$\text{let } n_0 = 1$$

$$5 + \frac{\log n}{c} + 2 + 5 \leq 15$$

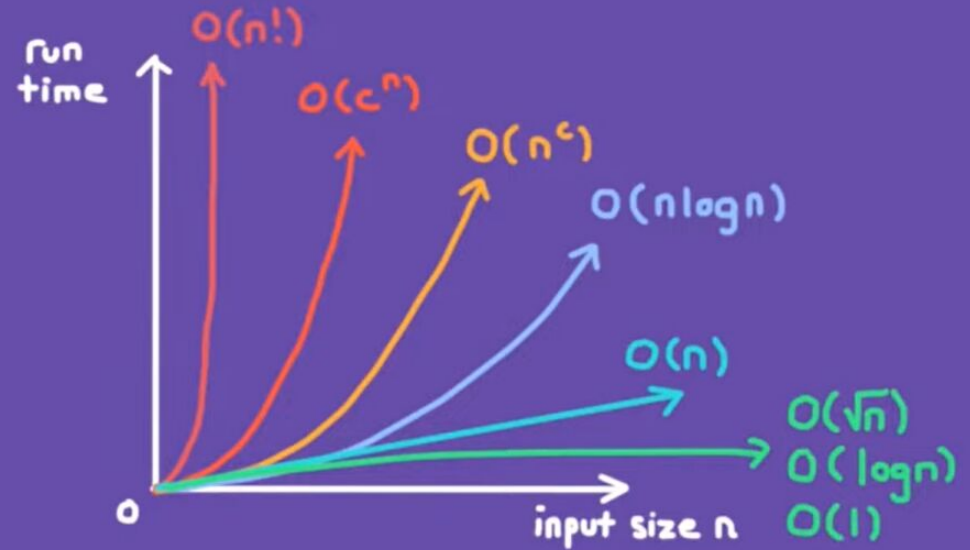
c

$$12 \leq 15 \quad \checkmark$$



n^n
 $n!$
 c^n
 n^c
 $+$
 $n \log n$
 n
 $\sqrt{n}$
 $\log n$
 c

Previously,
I said that
 $n \log n < n^2$
because $n \log n$
grows slower
than polynomial
functions



Prove that

$5n^2 + 3n \log n + 2n + 5$ is $O(n^2)$

$$5n^2 + 3n \log n + 2n + 5 \leq 15n^2$$

let $n_0 = 1$

$$5 + 0 + 2 + 5 \leq 15$$

$$12 \leq 15 \quad \checkmark$$

Therefore, $5n^2 + 3n \log n + 2n + 5$ is $O(n^2)$

since $5n^2 + 3n \log n + 2n + 5 \leq 15n^2$

for all $n \geq 1$ (by the definition of Big O, where $c=15$ and $n_0=1$)

$f(n)$ is $O(g(n))$

If there is a positive constant c
and a positive integer n_0

such that $0 \leq f(n) \leq c \cdot g(n)$
for all $n \geq n_0$

